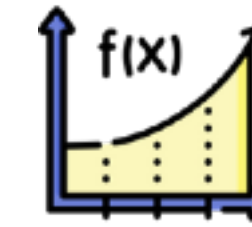
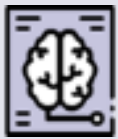


FUNCTIONS IN MACHINE LEARNING



- ▶ Disease Identification from X-rays: $f(\text{X-ray image}) = \text{disease diagnosis}$ 
- ▶ Treatment Recommendations: $f(\text{patient symptoms and history}) = \text{suggested treatments}$ 
- ▶ Epidemic Outbreak Prediction: $f(\text{recent health data}) = \text{risk of epidemic outbreak}$ 
- ▶ Predictive Health Monitoring: $f(\text{real-time health metrics}) = \text{immediate health risks}$ 
- ▶ Clinical Note Analysis: $f(\text{clinical notes}) = \text{disease diagnosis or treatment recommendation}$ 
- ▶ Drug Side Effect Identification: $f(\text{patient feedback}) = \text{potential drug side-effects}$ 
- ▶ Epidemic Surveillance from News: $f(\text{news articles}) = \text{potential epidemic outbreaks}$ 
- ▶ Brain MRI Analysis: $f(\text{MRI image}) = \text{presence of tumors or anomalies}$ 
- ▶ Cell Image Classification in Pathology: $f(\text{Cell image}) = \text{cell type and potential abnormalities}$ 

SUPERVISED LEARNING AND FUNCTIONS!

In supervised learning, we have a dataset consisting of input-output pairs, usually denoted as $(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$, where x is the input and y is the corresponding output. The goal is to learn a function f such that for a new input x_{new} , the function can predict the output y_{new} with accuracy ($y = f(x)$).

The idea of supervised learning is to learn a function f based on the training data and then test its performance on previously unseen data.